

Paper Practice 3.3**Linearity and Continuity**

WORKED KEY – Show all of your work in the space provided. Simplify before you decide – an equation can look nonlinear and simplify to a line.

- 1.** Determine whether each function is linear or nonlinear. Simplify first. (Simplify First)

a. $y = 4x^2 - (2x)^2 + 3x - 5$ ANSWER:
Linear. $(2x)^2 = 4x^2$, so $4x^2 - 4x^2 = 0$, leaving $y = 3x - 5$.

b. $y = 3x^3 - x^2 + 3x + 6$ ANSWER:
Nonlinear – the x^3 term does not cancel.

c. $4x - (2y)^2 = 3$ ANSWER: Nonlinear.
 $(2y)^2 = 4y^2$, and y is squared.

- 2.** Determine whether each function is linear or nonlinear. (Linear or Not)

a. $2x + 5y = 10$ ANSWER: Linear – already in standard form.

b. $y = x^2 - 4$ ANSWER: Nonlinear – x is squared.

c. $y = 7$ ANSWER: Linear – a constant function graphs as a horizontal line.

- 3.** Simplify, then decide whether the function is linear or nonlinear. (Simplify First)

a. $y = 6x^2 - (3x)^2 + 3x^2 + 4x + 1$
ANSWER: Linear. $6x^2 - 9x^2 + 3x^2 = 0$, leaving $y = 4x + 1$.

- 4.** Write each linear equation in standard form $ax + by = c$ with integer coefficients. (Standard Form)

a. $y = 3x - 8$ ANSWER: $3x - y = 8$

b. $y = -\frac{1}{2}x + 4$ ANSWER: $x + 2y = 8$
(multiply through by 2)

- 5.** Determine whether each situation is discrete or continuous. Explain. (Discrete or Continuous)

a. the number of tickets sold for a concert
ANSWER: Discrete – you cannot sell part of a ticket.

b. the temperature of a room over one hour
ANSWER: Continuous – temperature takes every value in between.

c. the number of students riding a bus
ANSWER: Discrete – students are counted in whole numbers.

- 6.** The Bargain Book Barn sells novels on a sliding scale: the more you buy, the cheaper each one is. Let $f(x)$ be the price for x books. Is $f(x)$ discrete or continuous? Explain. (Discrete or Continuous) ANSWER: Discrete – x can only be a whole number of books, so the graph is separate points.

- 7.** A function has domain $\{0, 1, 2, 3\}$ and range $\{2, 5, 8, 11\}$. Is it discrete or continuous? Explain.
(Discrete or Continuous) **ANSWER:** Discrete – the domain is a finite set of separated points.

- 8.** The table shows a function. Is it linear? Justify your answer using the rate of change. (Rate of Change)
- a.** x : 1, 2, 3, 4 y : 3, 6, 9, 12
ANSWER: Linear – y increases by 3 for each increase of 1 in x , a constant rate of change.

- 9.** The table shows a function. Is it linear? Justify your answer. (Rate of Change)
- a.** x : 1, 2, 3, 4 y : 2, 4, 8, 16
ANSWER: Nonlinear – the differences are 2, 4, 8. The rate of change is not constant.

- 10.** Give the domain and the range of the discrete function $\{(-2, 1), (0, 3), (2, 5)\}$. (Domain & Range) **ANSWER:** $D = \{-2, 0, 2\}$ $R = \{1, 3, 5\}$
- 11.** Error Analysis – Gus says $y = 4x^2 - (2x)^2 + 3x - 5$ is nonlinear because it contains x^2 . Identify his mistake and give the correct answer. (Error Analysis) **ANSWER:** He never simplified. $(2x)^2 = 4x^2$, so the x^2 terms cancel and $y = 3x - 5$, which is linear.
- 12.** Explain the difference between a discrete and a continuous function, then give a real-world example of each that is not used above. (Explain) **ANSWER:** Answers vary. Discrete = separated points, counted values; continuous = an unbroken line or curve, measured values.